

There exists a buyer and a seller

The seller has marginal cost of producing=0

The buyer has 2 valuation types which decides how he values the good giving his reservation value

$$b_i$$

With

$$b_H > b_L > 0$$

The buyer's payoff is

$$u_i(p, t) = \delta^t \cdot (b_i - p)$$

With $\delta \in (0,1)$

a)

function of indifference curves

$$\begin{aligned}\delta^t \cdot (b_i - p(t)) &= \bar{u} \\ \frac{d}{dt} \delta^t \cdot (b_i - p(t)) &= 0 \\ \log(\delta) \cdot \delta^t \cdot (b_i - p(t)) - \delta^t \cdot \frac{dp}{dt} &= 0 \\ \Leftrightarrow \frac{dp}{dt} &= (b_i - p(t)) \cdot \log(\delta)\end{aligned}$$

find indifference curves

$$\delta^t \cdot (b_i - p) = \bar{u} \Leftrightarrow \bar{u} \cdot \delta^{-t} = b_i - p \Leftrightarrow p(t) = b_i - \bar{u} \cdot \delta^{-t}$$

$$\frac{dp}{dt} = (b_i - p(t)) \cdot \log(\delta) = (b_i - (b_i - \bar{u} \cdot \delta^{-t})) \cdot \log(\delta) = \bar{u} \cdot \log(\delta) \cdot \delta^{-t}$$

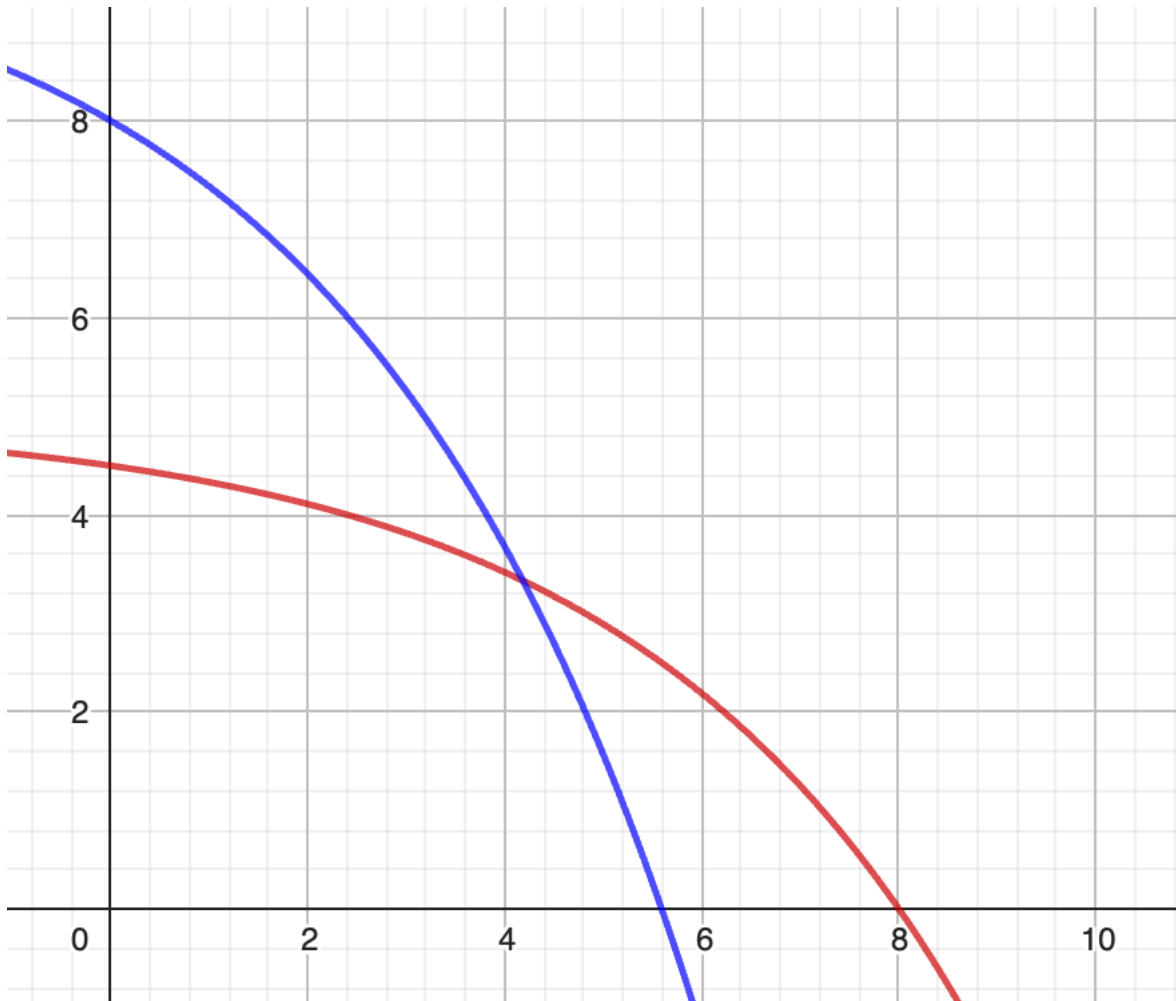
Thus

$$p'_H(t) = p'_L(t)$$

Then because $b_H > b_L$ we must have that the 2 can cross exactly once or not at all

b)

i plot in geogebra



Here i have

$$\delta = 0.75, b_H = 10, b_L = 5, \bar{u}_H = 2, \bar{u}_L = 0.5$$

We see that they cross exactly once

We note that utility is increasing down and to the left

c)

(ic) high type:

$$\delta^{t_L} \cdot (b_H - p_L) \leq \delta^{t_H} \cdot (b_H - p_H) \quad (1)$$

Low type:

$$\delta^{t_L} \cdot (b_L - p_L) \geq \delta^{t_H} \cdot (b_L - p_H) \quad (2)$$

d)

show

$$t_L = t_H \Leftrightarrow p_L = p_H$$

$$\begin{aligned}
t_L = t_H &\Rightarrow (1) \Leftrightarrow p_H \leq p_L \\
&\Rightarrow (2) \Leftrightarrow p_H \geq p_L \\
p_H = p_L &\Rightarrow (1) \Leftrightarrow t_L \geq t_H \\
&(2) \Rightarrow t_L \leq t_H
\end{aligned}$$

Thus in separating EQ we can never have equality in either

$$"t_L > t_H, p_L > p_H"$$

This is never a viable pooling EQ since this makes the high bundle strictly better for both parties

Both prefer low p and low t

In this instance L chooses H's bundle

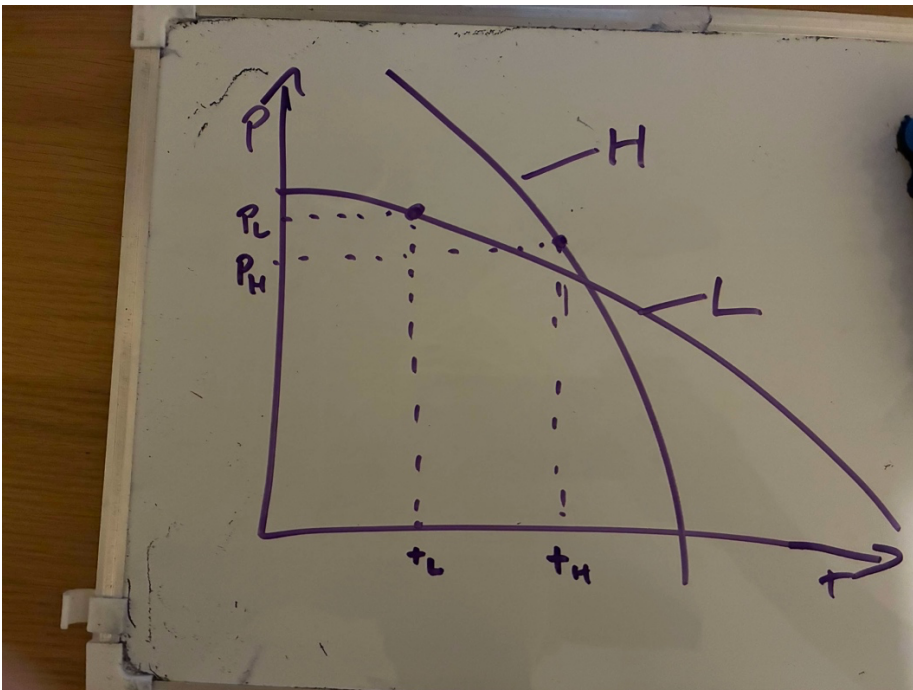
$$"t_L < t_H, p_L < p_H"$$

Same argument

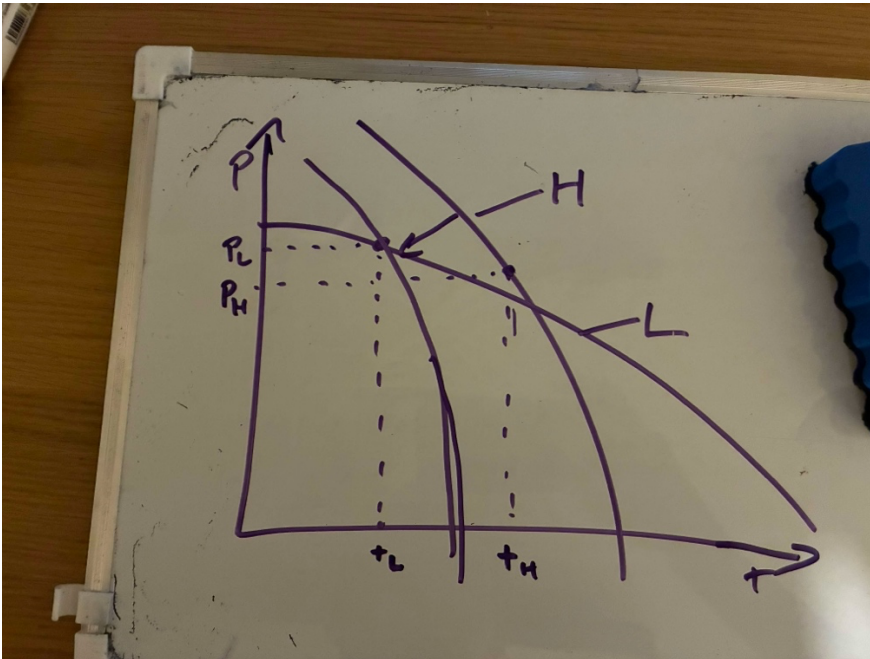
H chooses L's bundle

$$\text{AFC } t_L < t_H, p_L > p_H$$

We draw



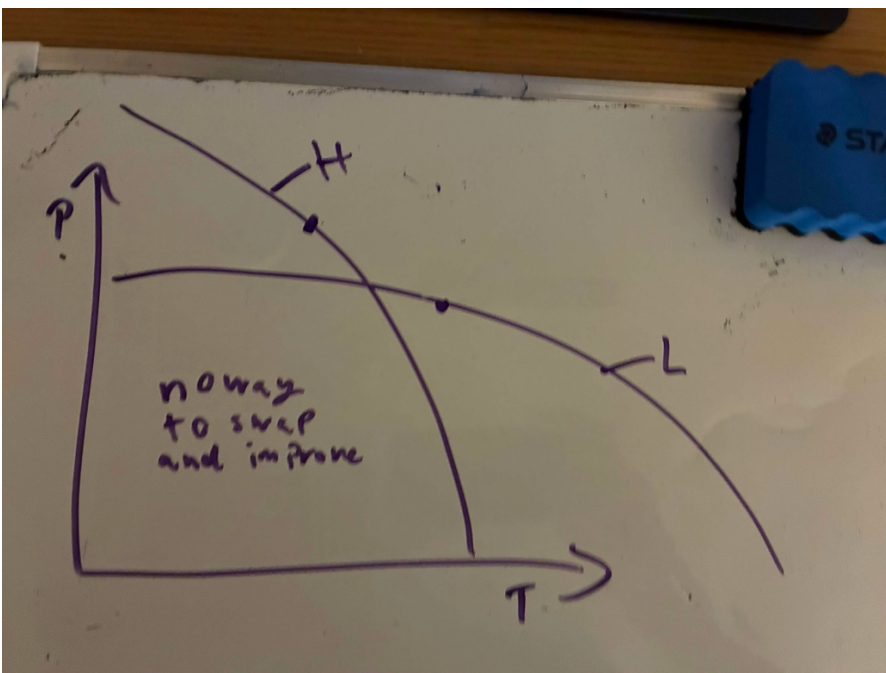
Seeing that if H takes the bundle meant for L he increases his utility



Thus this is not a separating EQ

Thus any separating EQ must satisfy

$$p_H > p_L, t_H < t_L$$



Thus this is might be a viable separating EQ

e)

assuming that everything is logable which probably isent so bad and if a somethin isent logable we know that there will either be no solutions or a constraint will never be binding

$$\delta^{t_L} \cdot \left(b_H - \frac{b_L}{2}\right) \leq \delta^0 \cdot \left(b_H - \frac{b_H}{2}\right) \Leftrightarrow t_L \leq \frac{\log\left(b_H - \frac{b_H}{2}\right) - \log\left(b_H - \frac{b_L}{2}\right)}{\log(\delta)}$$
$$\delta^{t_L} \cdot \left(b_L - \frac{b_L}{2}\right) \geq \delta^0 \cdot \left(b_L - \frac{b_H}{2}\right) \Leftrightarrow t_L \geq \frac{\log\left(b_L - \frac{b_H}{2}\right) - \log\left(b_L - \frac{b_L}{2}\right)}{\log(\delta)}$$